



Top 30 CAT Time Speed Distance Questions With Video Solutions

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Questions

Instructions

For the following questions answer them individually

Question 1

Two ants A and B start from a point P on a circle at the same time, with A moving clock-wise and B moving anti-clockwise. They meet for the first time at 10:00 am when A has covered 60% of the track. If A returns to P at 10:12 am, then B returns to P at

- A 10:25 am
- B 10:45 am
- C 10:18 am
- D 10:27 am

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Two ants A and B start from a point P on a circle at the same time, with A moving clock-wise and B moving anti-clockwise. They meet for the first time at 10:00 am when A has covered 60% of the track. If A returns to P at 10:12 am, then B returns to P at

10:12 AM

A $\rightarrow$ 60%
B $\rightarrow$ 40%
 $\frac{3}{2}$

60% in 12 m

40%

VIDEO SOLUTION

Question 2

A cyclist leaves A at 10 am and reaches B at 11 am. Starting from 10:01 am, every minute a motorcycle leaves A and moves towards B. Forty-five such motorcycles reach B by 11 am. All motorcycles have the same speed. If the cyclist had doubled his speed, how many motorcycles would have reached B by the time the cyclist reached B?

- A 22
- B 23
- C 15
- D 20

cracku

A cyclist leaves A at 10 am and reaches B at 11 am. Starting from 10:01 am, every minute a motorcycle leaves A and moves toward B. Forty-five such motorcycles reach B by 11 am. All motorcycles have the same speed. If the cyclist had doubled his speed, how many motorcycles would have reached B by the time the cyclist reached B?

10:30 am

VIDEO SOLUTION

Question 3

John jogs on track A at 6 kmph and Mary jogs on track B at 7.5 kmph. The total length of tracks A and B is 325 metres. While John makes 9 rounds of track A, Mary makes 5 rounds of track B. In how many seconds will Mary make one round of track A?

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John jogs on track A at 6 kmph and Mary jogs on track B at 7.5 kmph. The total length of tracks A and B is 325 metres. While John makes 9 rounds of track A, Mary makes 5 rounds of track B. In how many seconds will Mary make one round of track A?

$A+B=0.325 \text{ km}$

$J=6 \text{ km/h}$

$M=7.5 \text{ km/h}$

$\frac{9 \times A}{B} = \frac{5 \times B}{A}$

VIDEO SOLUTION

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Question 4

Two cars travel the same distance starting at 10:00 am and 11:00 am, respectively, on the same day. They reach their common destination at the same point of time. If the first car travelled for at least 6 hours, then the highest possible value of the percentage by which the speed of the second car could exceed that of the first car is

- A 20
- B 30
- C 25
- D 10

cracku

Two cars travel the same distance starting at 10:00 am and 11:00 am, respectively, on the same day. They reach their common destination at the same point of time. If the first car travelled for at least 6 hours, then the highest possible value of the percentage by which the speed of the second car could exceed that of the first car is

1 → 10 am = 6 + p
2 → 11 am = 5 + p

$\frac{D}{p+6} = \frac{D}{p+5}$

VIDEO SOLUTION

Question 5

The wheels of bicycles A and B have radii 30 cm and 40 cm, respectively. While traveling a certain distance, each wheel of A required 5000 more revolutions than each wheel of B. If bicycle B traveled this distance in 45 minutes, then its speed, in km per hour, was

- A 18π
- B 14π
- C 16π
- D 12π

cracku

The wheels of bicycles A and B have radii 30 cm and 40 cm, respectively. While traveling a certain distance, each wheel of A required 5000 more revolutions than each wheel of B. If bicycle B traveled this distance in 45 minutes, then its speed, in km per hour, was

$\frac{2\pi R_A}{C_A} = \frac{2\pi R_B}{C_B} + 5000$

$\frac{30}{C_A} = \frac{40}{C_B} + 5000$

$\frac{C_A}{C_B} = \frac{3}{4}$

VIDEO SOLUTION

Question 6

In a race of three horses, the first beat the second by 11 metres and the third by 90 metres. If the second beat the third by 80 metres, what was the length, in metres, of the racecourse?

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In a race of three horses, the first beat the second by 11 metres and the third by 90 metres. If the second beat the third by 80 metres, what was the length, in metres, of the racecourse?

A B C S

→ S S-11 S-90 S-80

VIDEO SOLUTION

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Question 7

One can use three different transports which move at 10, 20, and 30 kmph, respectively to reach from A to B. Amal took each mode of transport for $\frac{1}{3}^{\text{rd}}$ of his total journey time, while Bimal took each mode of transport for $\frac{1}{3}^{\text{rd}}$ of the total distance. The percentage by which Bimal's travel time exceeds Amal's travel time is nearest to

- A 22
- B 20
- C 19
- D 21

cracku

One can use three different transports which move at 10, 20, and 30 kmph, respectively, to reach from A to B. Amal took each mode of transport $\frac{1}{3}$ of his total journey time, while Bimal took each mode of transport $\frac{1}{3}$ of the total distance. The percentage by which Bimal's travel time exceeds Amal's travel time is nearest to

Amal Time: $\frac{10 \text{ km/h}}{T_{10}} + \frac{20 \text{ km/h}}{T_{20}} + \frac{30 \text{ km/h}}{T_{30}} = 60T$

Bimal Distance: $\frac{10 \text{ km/h}}{T_{10}} + \frac{20 \text{ km/h}}{T_{20}} + \frac{30 \text{ km/h}}{T_{30}} = 60T$

VIDEO SOLUTION

Question 8

Leaving home at the same time, Amal reaches the office at 10:15 am if he travels at 8 km/hr, and at 9:40 am if he travels at 15 km/hr. Leaving home at 9:10 am, at what speed, in km/hr, must he travel so as to reach office exactly at 10 am?

- A 13

- B 12
- C 14
- D 11

cracku

Leaving home at the same time, Aral reaches the office at 10:15 am if he travels at 8 km/hr, and at 9:40 am if he travels at 15 km/hr. Leaving home at 9:10 am, at what speed, in km/hr, must he travel so as to reach office exactly at 10 am?

$(10\frac{1}{4} - T) \times 8 = (9\frac{2}{3} - T) \times 15$

$82 - 8T = 80 - 5T$

$2 = 3T$

$T = \frac{2}{3}$

82 - 8T =

VIDEO SOLUTION

Question 9

A train travelled at one-thirds of its usual speed, and hence reached the destination 30 minutes after the scheduled time. On its return journey, the train initially travelled at its usual speed for 5 minutes but then stopped for 4 minutes for an emergency. The percentage by which the train must now increase its usual speed so as to reach the destination at the scheduled time, is nearest to

- A 50
- B 58
- C 67
- D 61

cracku

A train travelled at one-thirds of its usual speed, and hence reached the destination 30 minutes after the scheduled time. On its return journey, the train initially travelled at its usual speed for 5 minutes but then stopped for 4 minutes for an emergency. The percentage by which the train must now increase its usual speed so as to reach the destination at the scheduled time, is nearest to

$8 \times T = \frac{8}{3} \times (T + \frac{1}{2})$

$3T = T + \frac{1}{2}$

$T = \frac{1}{4} \text{ hr} = 15 \text{ min}$

$5 + \frac{1}{2} = 5.5$

$\frac{S}{3}$

$\frac{S}{12}$

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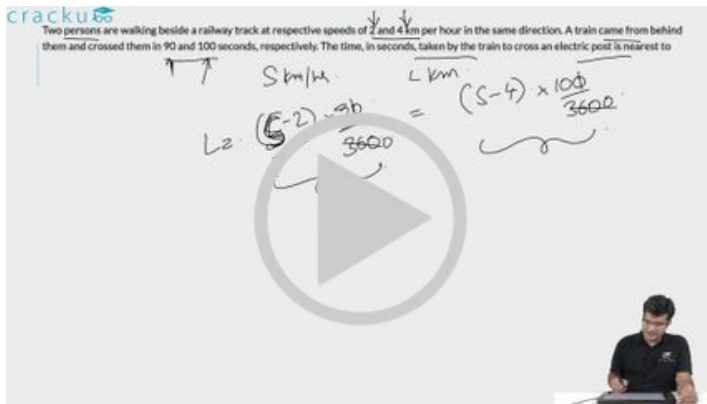
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Question 10

Two persons are walking beside a railway track at respective speeds of 2 and 4 km per hour in the same direction. A train came from behind them and crossed them in 90 and 100 seconds, respectively. The time, in seconds, taken by the train to cross an electric post is nearest to

- A 87
- B 82
- C 78
- D 75

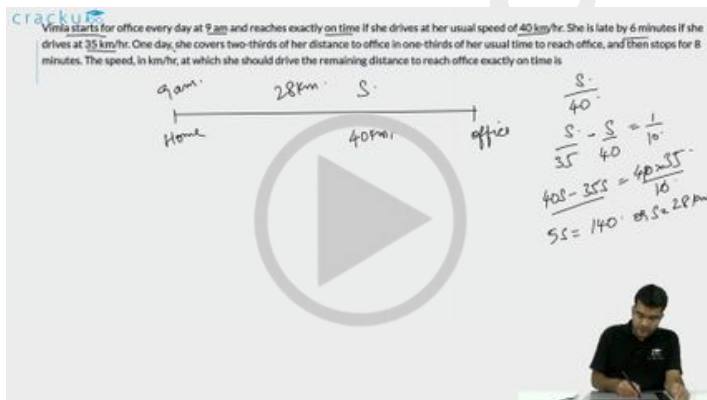


VIDEO SOLUTION

Question 11

Vimla starts for office every day at 9 am and reaches exactly on time if she drives at her usual speed of 40 km/hr. She is late by 6 minutes if she drives at 35 km/hr. One day, she covers two-thirds of her distance to office in one-third of her usual time to reach office, and then stops for 8 minutes. The speed, in km/hr, at which she should drive the remaining distance to reach office exactly on time is

- A 29
- B 26
- C 28
- D 27



VIDEO SOLUTION

Question 12

A and B are two railway stations 90 km apart. A train leaves A at 9:00 am, heading towards B at a speed of 40 km/hr. Another train leaves B at 10:30 am, heading towards A at a speed of 20 km/hr. The trains meet each other at

- A 11 : 45 am
- B 11 : 20 am
- C 11 : 00 am
- D 10 : 45 am

cracku

A and B are two railway stations 90 km apart. A train leaves A at 9:00 am, heading towards B at a speed of 40 km/hr. Another train leaves B at 10:30 am, heading towards A at a speed of 20 km/hr. The trains meet each other at

A) 11 : 45 am B) 11 : 20 am
C) 11 : 00 am D) 10 : 45 am

1.5x4

9:00am
40km/hr

90km

10:30am
20km/hr

VIDEO SOLUTION



CAT Syllabus PDF



Question 13

Anil, Sunil, and Ravi run along a circular path of length 3 km, starting from the same point at the same time, and going in the clockwise direction. If they run at speeds of 15 km/hr, 10 km/hr, and 8 km/hr, respectively, how much distance in km will Ravi have run when Anil and Sunil meet again for the first time at the starting point?

- A 4.8
- B 4.6
- C 5.2
- D 4.2

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Anil, Sunil, and Ravi run along a circular path of length 3 km, starting from the same point at the same time, and going in the clockwise direction. If they run at speeds of 15 km/hr, 10 km/hr, and 8 km/hr, respectively, how much distance in km will Ravi have run when Anil and Sunil meet again for the first time at the starting point?

3km

Anil = 15 km/hr $\Rightarrow \frac{3}{15} = 12 \text{ min}$
Sunil = 10 km/hr $\Rightarrow \frac{3}{10} = 18 \text{ min}$

Question 14

The distance from B to C is thrice that from A to B. Two trains travel from A to C via B. The speed of train 2 is double that of train 1 while traveling from A to B and their speeds are interchanged while traveling from B to C. The ratio of the time taken by train 1 to that taken by train 2 in travelling from A to C is

- A 5:7
- B 4:1
- C 1:4
- D 7:5

cracku

The distance from B to C is thrice that from A to B. Two trains travel from A to C via B. The speed of train 2 is double that of train 1 while traveling from A to B and their speeds are interchanged while traveling from B to C. The ratio of the time taken by train 1 to that taken by train 2 in travelling from A to C is

Train 1

Train 2

Question 15

In a car race, car A beats car B by 45 km. car B beats car C by 50 km. and car A beats car C by 90 km. The distance (in km) over which the race has been conducted is

- A 475
- B 450
- C 500
- D 550

cracku

In a car race, car A beats car B by 45 km. car B beats car C by 50 km. and car A beats car C by 90 km. The distance (in km) over which the race has been conducted is

A

B

A

B



CAT Formula PDF



Question 16

A and B are two points on a straight line. Ram runs from A to B while Rahim runs from B to A. After crossing each other. Ram and Rahim reach their destination in one minute and four minutes, respectively. if they start at the same time, then the ratio of Ram's speed to Rahim's speed is

- A $\frac{1}{2}$
- B $\sqrt{2}$
- C 2
- D $2\sqrt{2}$

cracku

A and B are two points on a straight line. Ram runs from A to B while Rahim runs from B to A. After crossing each other. Ram and Rahim reach their destination in one minute and four minutes, respectively. if they start at the same time, then the ratio of Ram's speed to Rahim's speed is

A) $\frac{1}{2}$
B) $\sqrt{2}$
C) 2
D) $2\sqrt{2}$

min. T.

Ace XT.
Bc = YT

Question 17

Two circular tracks T1 and T2 of radii 100 m and 20 m, respectively touch at a point A. Starting from A at the same time, Ram and Rahim are walking on track T1 and track T2 at speeds 15 km/hr and 5 km/hr respectively. The number of full rounds that Ram will make before he meets Rahim again for the first time is

- A 5
- B 3
- C 2
- D 4

cracku

Two circular tracks T₁ and T₂ of radii 100 m and 20 m, respectively touch at a point A. Starting from A at the same time, Ram and Rahim are walking on track T₁ and track T₂ at speeds 15 km/hr and 5 km/hr respectively. The number of full rounds that Ram will make before he meets Rahim again for the first time is

A) 5
C) 2

B) 3
D) 4

Handwritten calculations:
 $1 \text{ round} = \frac{2\pi \times 100}{15 \times \frac{5}{18}} = \frac{200}{3}$
 $2 \text{ rounds} = \frac{400}{3}$

VIDEO SOLUTION

Question 18

Two trains cross each other in 14 seconds when running in opposite directions along parallel tracks. The faster train is 160 m long and crosses a lamp post in 12 seconds. If the speed of the other train is 6 km/hr less than the faster one, its length, in m, is

- A 184
- B 192
- C 190
- D 180

cracku

Two trains cross each other in 14 seconds when running in opposite directions along parallel tracks. The faster train is 160 m long and crosses a lamp post in 12 seconds. If the speed of the other train is 6 km/hr less than the faster one, its length, in m, is

A) 184
C) 190

B) 192
D) 180

Handwritten calculations:
 $\frac{160}{12} = \frac{40}{3} \text{ m/s}$

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Question 19

Two trains A and B were moving in opposite directions, their speeds being in the ratio 5 : 3. The front end of A crossed the rear end of B 46 seconds after the front ends of the trains had crossed each other. It took another 69 seconds for the rear ends of the trains to cross each other. The ratio of length of train A to that of train B is

- A 3:2

B 5:3

C 2:3

D 2:1

cracku

Two trains A and B were moving in opposite directions, their speeds being in the ratio 5:3. The front end of A crossed the rear end of B 46 seconds after the front ends of the trains had crossed each other. It took another 69 seconds for the rear ends of the trains to cross each other. The ratio of length of train A to that of train B is

A) 3:2
C) 2:3

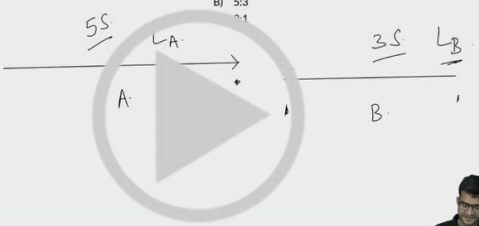
B) 5:3
D) 2:1

$L_B = 46x$

$5S$ L_A $3S$ L_B

A

B



VIDEO SOLUTION

Question 20

Mira and Amal walk along a circular track, starting from the same point at the same time. If they walk in the same direction, then in 45 minutes, Amal completes exactly 3 more rounds than Mira. If they walk in opposite directions, then they meet for the first time exactly after 3 minutes. The number of rounds Mira walks in one hour is

cracku

Mira and Amal walk along a circular track, starting from the same point at the same time. If they walk in the same direction, then in 45 minutes, Amal completes exactly 3 more rounds than Mira. If they walk in opposite directions, then they meet for the first time exactly after 3 minutes. The number of rounds Mira walks in one hour is

Answer: _____

$3M + 3A = C$
 $45A = C$

Circumference = C
Mira = M
Amal = A



VIDEO SOLUTION

Question 21

Trains A and B start traveling at the same time towards each other with constant speeds from stations X and Y, respectively. Train A reaches station Y in 10 minutes while train B takes 9 minutes to reach station X after meeting train A. Then the total time taken, in minutes, by train B to travel from station Y to station X is

A 6

B 15

C 10

D 12

Trains A and B start traveling at the same time towards each other with constant speeds from stations X and Y, respectively. Train A reaches station Y in 10 minutes while train B takes 9 minutes to reach station X after meeting train A. Then the total time taken, in minutes, by train B to travel from station Y to station X is

A) 6
B) 15
C) 10

cracku

VIDEO SOLUTION

CAT Percentile Predictor



Question 22

Two ships meet mid-ocean, and then, one ship goes south and the other ship goes west, both travelling at constant speeds. Two hours later, they are 60 km apart. If the speed of one of the ships is 6 km per hour more than the other one, then the speed, in km per hour, of the slower ship is

- A 20
- B 12
- C 18
- D 24

Two ships meet mid-ocean, and then, one ship goes south and the other ship goes west, both travelling at constant speeds. Two hours later, they are 60 km apart. If the speed of one of the ships is 6 km per hour more than the other one, then the speed, in km per hour, of the slower ship is

A) 20
B) 12
C) 18
D) 24

cracku

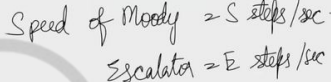
VIDEO SOLUTION

Question 23

Two cars travel from different locations at constant speeds. To meet each other after starting at the same time, they take 1.5 hours if they travel towards each other, but 10.5 hours if they travel in the same direction. If the speed of the slower car is 60 km/hr, then the distance traveled, in km, by the slower car when it meets the other car while traveling towards each other, is

- A 100

- D** 150

$$\begin{aligned} A &> B \\ \frac{S}{A+B} &= 1.5 \\ \frac{S}{A-B} &= 10.5 \\ \frac{A+B}{A-B} &= 7 \\ A+B &= 7A-7B \\ 6A &= 8B \end{aligned}$$


B 12

C 8

D 6

Two ships are approaching a port along straight routes at constant speeds. Initially, the two ships and the port formed an equilateral triangle with sides of length 24 km. When the slower ship travelled 8 km, the triangle formed by the new positions of the two ships and the port became right-angled. When the faster ship reaches the port, the distance, in km, between the other ship and the port will be

A) 4
B) 12
C) 8

The diagram shows an equilateral triangle with side length 24 km. The vertices are labeled 'Ship 1' (bottom left), 'Ship 2' (bottom right), and 'Port' (top). Ship 1 moves 8 km towards the Port, leaving a segment of 16 km. A new point is marked on the line between Ship 1 and the Port. A line segment connects this new point to Ship 2, and a right-angle symbol is shown at this intersection. A large, faint watermark of a play button is visible in the background of the diagram.

 VIDEO SOLUTION

Question 26

The minor angle between the hours hand and minutes hand of a clock was observed at 8:48 am. The minimum duration, in minutes, after 8.48 am when this angle increases by 50% is

- | | |
|----------|-----------------|
| A | $\frac{36}{11}$ |
| B | 2 |
| C | 4 |
| D | $\frac{24}{11}$ |

The minor angle between the hours hand and minutes hand of a clock was observed at 8:48 am. The minimum duration, in minutes, after 8:48 am when this angle increases by 50% is

A) $\frac{36}{11}$
B) 2
C) 4
D) $\frac{24}{11}$

minutes

Minute Hand = 6°

Hour Hand = $\frac{1}{2}^\circ$

60°

240°

 VIDEO SOLUTION

Question 27

Brishti went on an 8-hour trip in a car. Before the trip, the car had travelled a total of x km till then, where x is a whole number and is palindromic, i.e., x remains unchanged when its digits are reversed. At the end of the trip, the car had travelled a total of 26862 km till then, this number again being palindromic. If Brishti never drove at more than 110 km/h, then the greatest possible average speed at which she drove during the trip, in km/h, was

- A** 110
- B** 90

C 100

D 80

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Brishti went on an 8-hour trip in a car. Before the trip, the car had travelled a total of x km till then, where x is a whole number and is palindromic, i.e., x remains unchanged when its digits are reversed. At the end of the trip, the car had travelled a total of 26862 km till then, this number again being palindromic. If Brishti never drove at more than 110 km/h, then the greatest possible average speed at which she drove during the trip, in km/h, was

A) 110
B) 100
C) 100
D) 80

26862
- 25982

800

$800 / 8 = 100$

VIDEO SOLUTION

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Question 28

Arvind travels from town A to town B, and Surbhi from town B to town A, both starting at the same time along the same route. After meeting each other, Arvind takes 6 hours to reach town B while Surbhi takes 24 hours to reach town A. If Arvind travelled at a speed of 54 km/h, then the distance, in km, between town A and town B is

cracku

Arvind travels from town A to town B, and Surbhi from town B to town A, both starting at the same time along the same route. After meeting each other, Arvind takes 6 hours to reach town B while Surbhi takes 24 hours to reach town A. If Arvind travelled at a speed of 54 km/h, then the distance, in km, between town A and town B is

Answer: _____

Arvind = 54 km/h

Speed A = $\frac{24}{6} = 4$

Speed B = $\frac{24}{6} = 4$

Arbhi = 27

Arvind = 54 km/h

Speed A = $\frac{24}{6} = 4$

Speed B = $\frac{24}{6} = 4$

Arbhi = 27

VIDEO SOLUTION

Question 29

A boat takes 2 hours to travel downstream a river from port A to port B, and 3 hours to return to port A. Another boat takes a total of 6 hours to travel from port B to port A and return to port B. If the speeds of the boats and the river are constant, then the time, in hours, taken by the slower boat to travel from port A to port B is

A $12(\sqrt{5} - 2)$

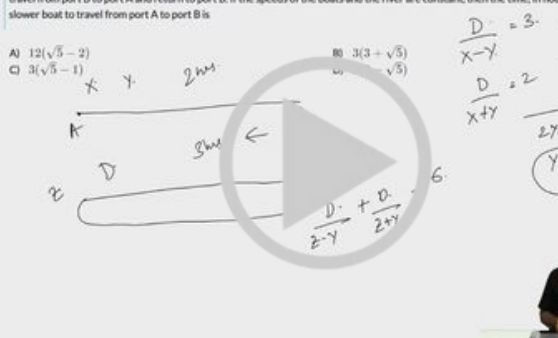
B $3(3 + \sqrt{5})$

C $3(\sqrt{5} - 1)$

D $3(3 - \sqrt{5})$

A boat takes 2 hours to travel downstream a river from port A to port B, and 3 hours to return to port A. Another boat takes a total of 6 hours to travel from port B to port A and return to port B. If the speeds of the boats and the river are constant, then the time, in hours, taken by the slower boat to travel from port A to port B is

A) $12(\sqrt{5} - 2)$
 B) $3(\sqrt{5} - 1)$



VIDEO SOLUTION

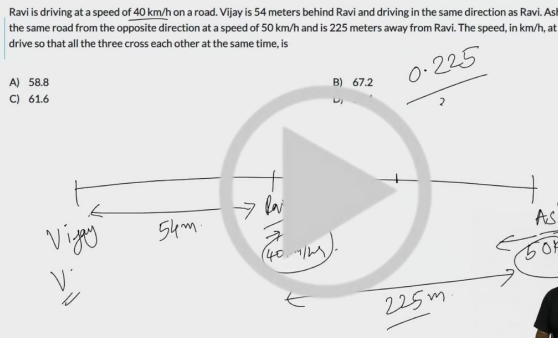
Question 30

Ravi is driving at a speed of 40 km/h on a road. Vijay is 54 meters behind Ravi and driving in the same direction as Ravi. Ashok is driving along the same road from the opposite direction at a speed of 50 km/h and is 225 meters away from Ravi. The speed, in km/h, at which Vijay should drive so that all the three cross each other at the same time, is

- A 58.8
- B 67.2
- C 61.6
- D 64.4

Ravi is driving at a speed of 40 km/h on a road. Vijay is 54 meters behind Ravi and driving in the same direction as Ravi. Ashok is driving along the same road from the opposite direction at a speed of 50 km/h and is 225 meters away from Ravi. The speed, in km/h, at which Vijay should drive so that all the three cross each other at the same time, is

A) 58.8
 B) 67.2
 C) 61.6



VIDEO SOLUTION

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Answers

1.D	2.C	3.48	4.A	5.C	6.880	7.A	8.B
9.C	10.B	11.C	12.C	13.A	14.A	15.B	16.C
17.B	18.C	19.A	20.8	21.B	22.C	23.B	24.60
25.B	26.D	27.C	28.972	29.D	30.C		

Explanations

1.D

When A and B met for the first time at 10:00 AM, A covered 60% of the track.

So B must have covered 40% of the track.

It is given that A returns to P at 10:12 AM i.e A covers 40% of the track in 12 minutes

60% of the track in 18 minutes

B covers 40% of track when A covers 60% of the track.

B covers 40% of the track in 18 minutes.

B will cover the rest 60% in 27 minutes, hence it will return to B at 10:27 AM

 VIDEO SOLUTION

2.C

It is given that starting from 10:01 am, every minute a motorcycle leaves A and moves towards B.

Forty-five such motorcycles reach B by 11 am.

It means that the forty-fifth motorcycle starts at 10:45 AM at A and reaches B by 11:00 AM i.e 15 minutes.

Since the speed of all the motorcycles is the same, all the motorcycles will take the same duration i.e 15 minutes.

If the cyclist doubles the speed, then he will reach B by 10:30 AM. (Since if the speed is doubled, time is reduced by half)

Since each motorcycle takes 15 minutes to reach B, 15 motorcycles would have reached B by the time the cyclist reaches B

 VIDEO SOLUTION

3.48

Speed of John = 6kmph

Speed of Mary = 7.5 kmph

Lengths of tracks A and B = 325 m

Let the length of track A be a, then the length of track B = 325-a

9 rounds of John on track A = 5 rounds of Mary on track B

$$\frac{9 \times a}{6 \times \frac{5}{18}} = \frac{5 \cdot (325 - a)}{7.5 \times \frac{5}{18}}$$

On solving we get , 13a=1300

a=100

The length of track A = 100m, track B = 225m

Mary makes one round of track A = $\frac{100}{7.5 \times \frac{5}{18}}$

= 48 sec

 VIDEO SOLUTION

4. A

Let the speed of cars be a and b and the distance = d

Minimum time taken by 1st car = 6 hours,

For maximum difference in time taken by both of them, car 1 has to start at 10:00 AM and car 2 has to start at 11:00 AM.

Hence, car 2 will take 5 hours.

Hence $a = \frac{d}{6}$ and $b = \frac{d}{5}$

Hence the speed of car 2 will exceed the speed of car 1 by $\frac{\frac{d}{5} - \frac{d}{6}}{\frac{d}{6}} \times 100 = \frac{\frac{d}{30}}{\frac{d}{6}} \times 100 = 20$

 VIDEO SOLUTION

5. C

Distance covered by A in 1 revolution = $2\pi \times 30 = 60\pi$

Distance covered by B in 1 revolution = $2\pi \times 40 = 80\pi$

Now, $(5000+n)60\pi = 80\pi n$

$\Rightarrow 15000 = 4n - 3n \Rightarrow n = 15000$

Then distance travelled by B = $15000 \times 80\pi$ cm = 12π km

Hence, the speed = $\frac{12\pi \times 60}{45} = 16\pi$

 VIDEO SOLUTION



CAT Daily Targets

Step by step, heading towards 99+ percentile




6. 880

Assuming the length of race course = x and the speed of three horses be a, b and c respectively.

Hence, $\frac{x}{a} = \frac{x-11}{b}$ (1)

and $\frac{x}{a} = \frac{x-90}{c}$ (2)

Also, $\frac{x}{b} = \frac{x-80}{c}$ (3)

From 1 and 2, we get, $\frac{x-11}{b} = \frac{x-90}{c}$ (4)

Dividing (3) by (4), we get, $\frac{x-11}{x} = \frac{x-90}{x-80}$

$\Rightarrow (x-11)(x-80) = x(x-90)$

$\Rightarrow 91x - 90x = 880 \Rightarrow x = 880$

 VIDEO SOLUTION

7. A

Assume the total distance between A and B as d and time taken by Amal = t

Since Amal travelled $\frac{1}{3}^{rd}$ of his total journey time in different speeds

$$d = \frac{t}{3} \times 10 + \frac{t}{3} \times 20 + \frac{t}{3} \times 30 = 20t$$

$$\text{Total time taken by Bimal} = \frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3}$$

$$= \frac{20t}{3} \times \frac{1}{10} + \frac{20t}{3} \times \frac{1}{20} + \frac{20t}{3} \times \frac{1}{30} = \frac{20t(6+3+2)}{3 \times 30} = \frac{11}{9}t$$

Hence, the ratio of time taken by Bimal to time taken by Amal = $\frac{\frac{11}{9}t}{t} = \frac{11}{9}$

Therefore, Bimal will exceed Amal's time by $\frac{\frac{11}{9}t - t}{t} \times 100 = 22.22$

[VIDEO SOLUTION](#)

8. B

The difference in the time take to traverse the same distance 'd' at two different speeds is 35 minutes.

$$\text{Equating this: } \frac{d}{8} - \frac{d}{15} = \frac{35}{60}$$

On solving, we obtain $d = 10 \text{ kms}$. Let $x \text{ kmph}$ be the speed at which Amal needs to travel to reach the office in 50 minutes; then

$$\frac{10}{x} = \frac{50}{60} \text{ or } x = 12 \text{ kmph. Hence, Option B is the correct answer.}$$

[VIDEO SOLUTION](#)

9. C

Let the total distance be 'D' km and the speed of the train be 's' kmph. The time taken to cover D at speed d is 't' hours. Based on the information: on equating the distance, we get $s \times t = \frac{s}{3} \times (t + \frac{1}{2})$

On solving we acquire the value of $t = \frac{1}{4}$ or 15 mins. We understand that during the return journey, the first 5 minutes are spent traveling at speed 's' {distance traveled in terms of $s = \frac{s}{12}$ }. Remaining distance in terms of 's' = $\frac{s}{4} - \frac{s}{12} = \frac{s}{6}$

The rest 4 minutes of stoppage added to this initial 5 minutes amounts to a total of 9 minutes. The train has to complete the rest of the journey in $15 - 9 = 6 \text{ mins}$ or $\{1/10 \text{ hours}\}$. Thus, let 'x' kmph be the new value of speed. Based on the above, we get $\frac{s}{x} = \frac{1}{10} \text{ or } x = \frac{10s}{6}$

Since the increase in speed needs to be calculated: $\frac{(\frac{10s}{6} - s)}{s} \times 100 = \frac{200}{3} \approx 67\%$ increase.

Hence, Option C is the correct answer.

[VIDEO SOLUTION](#)

10. B

Let the length of the train be $l \text{ kms}$ and speed be $s \text{ kmph}$. Base on the two scenarios presented, we obtain:

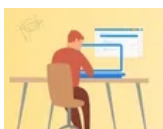
$$\frac{l}{s-2} = \frac{90}{3600} \dots (i) \text{ and } \frac{l}{s-4} = \frac{100}{3600} \dots (ii)$$

On dividing (ii) by (i) and simplifying we acquire the value of s as 22 kmph . Substituting this value in (i), we have $l = \frac{90}{3600} \times 20 \text{ kms}$ {keeping it in km and hours for convenience}

Since we need to find $\frac{l}{s}$, let this be equal to x . Then, $x = 90 \times \frac{20}{22} = 81.81 \approx 82 \text{ seconds}$

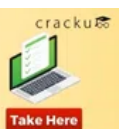
Hence, Option B is the correct choice.

[VIDEO SOLUTION](#)



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CAT Sectional Tests



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11. C

Let distance = d

$$\text{Given, } \frac{d}{35} - \frac{d}{40} = \frac{6}{60}$$

$$\Rightarrow d = 28\text{km}$$

The actual time taken to travel 28km = $28/40 = 7/10$ hours = 42 min.

Given time taken to travel $58/3$ km = $1/3 * 42 = 14$ min.

Then a break of 8 min.

To reach on time, he should cover remaining $28/3$ km in 20 min $\Rightarrow$ Speed = $\frac{(\frac{28}{3})}{\frac{20}{60}} = 28$ km/hr

 VIDEO SOLUTION

12. C

The distance travelled by A between 9:00 Am and 10:30 Am is $3/2 * 40 = 60$ km.

Now they are separated by 30 km

Let the time taken to meet = t

Distance travelled by A in time t + Distance travelled by B in time t = 30

$$40t + 20t = 30 \Rightarrow t = 1/2 \text{ hour}$$

Hence they meet at 11:00 AM

 VIDEO SOLUTION

13. A

Anil and Sunil will meet at a first point after LCM $(\frac{3}{15}, \frac{3}{10}) = 3/5$ hr

In the mean time, distance travelled by ravi = $8 * 3/5 = 4.8$ km

 VIDEO SOLUTION

14. A

Let the distance from A to B be "x", then the distance from B to C will be 3x. Now the speed of Train 2 is double of Train 1. Let the speed of Train 1 be "v", then the speed of Train 2 will be "2v" while travelling from A to B.

Time taken by Train 1 = (x/v)

Time taken by Train 2 = $(x/2v)$

Now from B to C distance is "3x" and the speed of Train 2 is (v) and the speed of Train 1 is (2v).

Time taken by Train 1 = $3x/2v$

Time taken by Train 2 = $3x/v$

Total time taken by Train 1 = $x/v(1+(3/2)) = (5/2)(x/v)$

Total time taken by Train 2 = $x/v(3+(1/2)) = (7/2)(x/v)$

$$\text{Ratio of time taken} = \frac{5}{2} = \frac{5}{7}$$

 VIDEO SOLUTION

15. B

Now car A beats car B by 45km. Let the speed of car A be v(a) and speed of car B be v(b).

$$\frac{v(a)}{v(b)} = \frac{m}{m-45} \dots\dots(1) \text{ where "m" is the entire distance of the race track.}$$

Moreover $\frac{v(b)}{v(c)} = \frac{m}{m-50} \dots\dots(2)$

and finally $\frac{v(a)}{v(c)} = \frac{m}{m-90} \dots\dots(3)$

Multiplying (1) and (2) we get (3). $\frac{m}{m-90} = \frac{m}{m-45} \left(\frac{m}{m-50} \right)$

Solving we get $m=450$ which is the length of the entire race track

[▶ VIDEO SOLUTION](#)



16. C

Let the speed of Ram be $v(r)$ and the speed of Rahim be $v(h)$ respectively. Let them meet after time "t" from the beginning.

Hence Ram will cover $v(r)(t)$ during that time and Rahim will cover $v(h)t$ respectively.

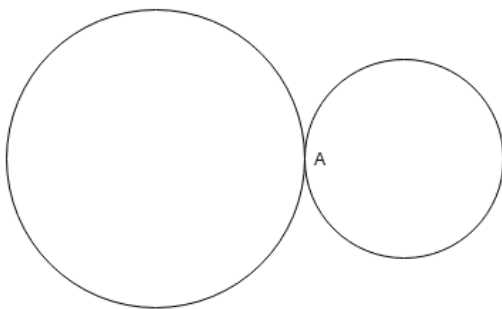
Now after meeting Ram reaches his destination in 1 min i.e. Ram covered $v(h)t$ in 1 minute or $v(r)(1) = v(h)(t)$

Similarly Rahim reaches his destination in 4 min i.e. Rahim covered $v(r)t$ in 4 minutes or $v(h)(4) = v(r)(t)$

Dividing both the equations we get $\frac{v(r)}{4v(h)} = \frac{v(h)}{v(r)}$ or $\frac{v(r)}{v(h)} = 2$ Hence the ratio is 2.

[▶ VIDEO SOLUTION](#)

17. B



To complete one round Ram takes 100m/15kmph and Rahim takes 20m/5kmph

They meet for the first time after L.C.M of (100m/15kmph , 20m/5kmph) = 100m/5kmph=20m/kmph.

Distance traveled by Ram =20m/kmph * 15kmph =300m.

So, he must have ran 300/100=3 rounds.

Note:

CAT gave both 2 and 3 as correct answers because of the word '**before**'.

[▶ VIDEO SOLUTION](#)

18. C

Speed of the faster train = $\frac{160}{12} = \frac{40}{3}$ m/s

Speed of the slower train = $\frac{40}{3} - \left(6 \times \frac{5}{18} \right) = \frac{35}{3}$ m/s

Sum of speeds (when the trains travel towards each other) = $\frac{40}{3} + \frac{35}{3} = 25$ m/s

Let the slower train be x metres long; then: $\frac{160+x}{25} = 14$

On solving, $x = 190 \text{ m}$

 VIDEO SOLUTION

19. A

Considering the length of train A = L_a , length of train B = L_b .

The speed of train A be $5x$, speed of train B be $3x$.

From the information provided :

The front end of A crossed the rear end of B 46 seconds after the front ends of the trains had crossed each other.

In this case, train A traveled a distance equivalent to the length of train B which is L_b at a speed of $5x+3x = 8x$ because both the trains are traveling in the opposite direction.

Hence $(8x)(46) = L_b$.

In the information provided :

It took another 69 seconds for the rear ends of the trains to cross each other.

In the next 69 seconds

The train B traveled a distance equivalent to the length of train A in this 69 seconds.

Hence $(8x)(69) = L_a$.

$L_a/L_b = 69/46 = 3/2 = 3 : 2$

 VIDEO SOLUTION

20. 8

Considering the distance travelled by Mira in one minute = M ,

The distance traveled by Amal in one minute = A .

Given if they walk in the opposite direction it takes 3 minutes for both of them to meet. Hence $3(A+M) = C$. (1)

C is the circumference of the circle.

Similarly, it is mentioned that if both of them walk in the same direction Amal completes 3 more rounds than Mira :

Hence $45(A-M) = 3C$. (2)

Multiplying (1)*15 we have :

$45A + 45M = 15C$.

$45A - 45M = 3C$.

Adding the two we have $A = \frac{18C}{90}$

Subtracting the two $M = \frac{12C}{90}$

Since Mira travels $\frac{12C}{90}$ in one minute, in one hour she travels : $\frac{12C}{90} \cdot 60 = 8C$

Hence a total of 8 rounds.

Alternatively,

Let the length of track be L

and velocity of Mira be a and Amal be b

Now when they meet after 45 minutes Amal completes 3 more rounds than Mira

so we can say they met for the 3rd time moving in the same direction

so we can say they met for the first time after 15 minutes

So we know Time to meet = Relative distance / Relative velocity

so we get $\frac{15}{60} = \frac{L}{a-b}$ (1)

Now When they move in opposite direction

They meet after 3 minutes

$$\text{so we get } \frac{3}{60} = \frac{L}{a+b} \quad (2)$$

Dividing (1) and (2)

$$\text{we get } \frac{(a+b)}{(a-b)} = 5$$

$$\text{or } 4a = 6b$$

$$\text{or } a = 3b/2$$

Now substituting in (1)

we get :

$$\frac{L}{b} \times 2 = \frac{15}{60}$$

$$\text{so } \frac{L}{b} = \frac{1}{8}$$

So we can say 1 round is covered in $\frac{1}{8}$ hours

so in 1-hour total rounds covered = 8.

 VIDEO SOLUTION



21. B



M - First meeting point

Let the speeds of trains A and B be 'a' and 'b', respectively.

$$\frac{x}{a} = \frac{D-x}{b}$$

It is given,

$$\frac{D}{a} = 10 \text{ and } \frac{x}{b} = 9$$

$$\frac{x}{\frac{D}{10}} = \frac{\frac{D-x}{9}}{\frac{x}{9}}$$

$$\frac{10x}{D} = \frac{9D-9x}{x}$$

$$10x^2 = 9D^2 - 9Dx$$

$$10x^2 + 9Dx - 9D^2 = 0$$

$$\text{Solving, we get } x = \frac{3D}{5}$$

$$\frac{x}{b} = 9$$

$$\frac{3D}{b \times 5} = 9$$

$$\frac{D}{b} = 15$$

The total time taken by train B to travel from station Y to station X is 15 minutes.

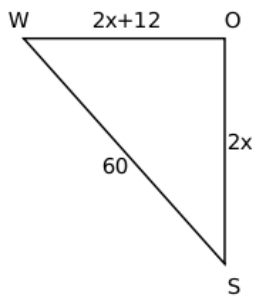
The answer is option B

 VIDEO SOLUTION

22. C

Let the speeds of two ships be 'x' and 'x+6' km per hour

Distance covered in 2 hours will be 2x and 2x+12



It is given,

$$(2x)^2 + (2x + 12)^2 = 60^2$$

$$(x)^2 + (x + 6)^2 = 30^2$$

$$2x^2 + 12x + 36 = 900$$

$$x^2 + 6x + 18 = 450$$

$$x^2 + 6x - 432 = 0$$

Solving, we get $x = 18$

The speed of slower ship is 18 kmph

The answer is option C.

VIDEO SOLUTION

23. B

Both the cars take 1.5 hrs to meet when they travel towards each other.

It is given, speed of slower car is 60 km/hr

Therefore, distance covered by slower car before they meet = $60 \times 1.5 = 90$ km

The answer is option B.

VIDEO SOLUTION

24. 60

Let the speed of Moody be 'x' steps/sec and that of the escalator be 'y' steps/sec.

In 30 seconds, Moody will finish riding the escalator when going in the same direction.

Thus, total steps = $30(x+y)$

If Moody's speed becomes twice, the time becomes 20 seconds.

Thus, total steps = $20(2x+y)$

$$\text{Or } 30x + 30y = 40x + 20y$$

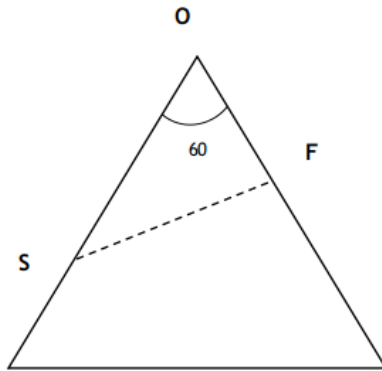
$$\text{Or } x = y$$

So, total steps = $60y$.

Time taken by only escalator = $60y/y = 60$ s.

VIDEO SOLUTION

25. B



Let S be the slower ship and F be the faster ship.

It is given that when S travelled 8 km, the positions of ships with the port is forming a right triangle.

Since one of the angles is 60 (since one vertex is still part of the equilateral triangle), the other two vertices will have angles of 30 and 90.

The distance between O and S = $24 - 8 = 16$

In triangle OFS, $\cos 60^\circ = \frac{OF}{OS}$

Thus, $OF = 8$.

Thus in the time, S covered 8 km, F will cover $24 - 8 = 16$ km.

Thus, the ratio of their speeds is 2:1,

Thus, when F covers 24 km, S will cover 12 km.

The correct option is B.

[VIDEO SOLUTION](#)

100+ CAT Quant Questions



26. D

The given time is 8:48 AM.

Angle made by hours hand w.r.t 12 is $8 * 30$ (30 degrees in 1 hour) + $0.5 * 48$ (0.5 degree in 1 minute) = $240 + 24 = 264$ degrees.

Angle made by minutes hands w.r.t 12 is $48 * 6 = 288$ degrees.

=> The angle between them is $288 - 264 = 24$ degrees.

This should further increase by 12 degrees (50% of 24)

After m minutes, the further increase in angle = $(6 - 0.5)m = \frac{11}{2}m = 12 \Rightarrow m = \frac{24}{11}$

[VIDEO SOLUTION](#)

27. C

Given the total number of kilometres travelled, including the trip = is 26862 Km, and the duration of the trip is 8 hrs.

If avg. speed of the car during the trip is 's' => the km travelled till just before the trip is $26862 - 8s$, which should also be a palindrome.

=> From the options if $s = 110 \Rightarrow$ The reading will be $26862 - 110 * 8 = 25982$ (Not a palindrome)

=> If $s = 100$ => The reading will be $26862 - 100 \times 8 = 26062$ => It is a palindrome.

=> $s = 100$ is the correct option.

 VIDEO SOLUTION

28. 972

Let us assume the speeds of Arvind and Surbhi are 'a' and 's', respectively.

Let us say they meet after 't' hours

=> Arvind travelled $s \times t$ distance in 6 hrs and Surbhi travelled $a \times t$ in 24 hrs

=> $s \times t = a \times 6$ and $a \times t = s \times 24$ => $t^2 = 6 \times 24$ => $t = 12$

Given $a = 54$ => $s \times 12 = 54 \times 6$ => $s = 27$.

=> Total distance between A and B is $(s+a) \times t = (54+27) \times 12 = 81 \times 12 = 972$ Kms.

 VIDEO SOLUTION

29. D

Let us assume the speed of the 1st boat is b, the 2nd boat is s, and the river's speed is r.

Let 'd' be the distance between A and B.

=> $d = 2(b+r)$ and $d = 3(b-r)$

=> $b + r = d/2$ and $b - r = d/3$ => $r = d/12$ (subtracting both equations).

Now, it is given that

$$\frac{d}{s+r} + \frac{d}{s-r} = 6$$

$$\Rightarrow \frac{d}{s + \frac{d}{12}} + \frac{d}{s - \frac{d}{12}} = 6$$

$$\Rightarrow 2ds = 6 \left(s^2 - \frac{d^2}{144} \right)$$

$$\Rightarrow 144s^2 - 48ds - d^2 = 0$$

Solving the quadratic equation, we get:

$$s = d \left(\frac{(48 + \sqrt{48^2 + 4(144)})}{2 \times 144} \right)$$

$$s = d \left(\frac{1}{6} + \frac{\sqrt{5}}{12} \right)$$

=> Required value of $\frac{d}{s+r}$

$$= \frac{d}{\frac{d}{6} + \frac{\sqrt{5}d}{12} + \frac{d}{12}}$$

$$= \frac{12}{3 + \sqrt{5}} = \frac{(12)(3 - \sqrt{5})}{4}$$

$$= 3(3 - \sqrt{5})$$

 VIDEO SOLUTION

30. C

It is given that the speed of Ravi is 40 kmph, which is equal to $\frac{100}{9}$ m/s. It is also known that the speed of Ashok is 50 kmph, which is equal to $\frac{125}{9}$ m/s.

It is known that the distance between Ravi and Ashok is 225 meters, and the relative speed of Ravi and Ashok is $\frac{125}{9} + \frac{100}{9} = 25$ m/s

Hence, they will meet each other in $\frac{225}{25} = 9$ seconds. The distance traveled by Ravi in these 9 seconds is $\frac{100}{9} \times 9 = 100$ meters.

Since Vijay was already 54 meters behind Ravi when they were starting, Vijay must travel $(100+54) = 154$ meters in these 9 seconds.

Hence, the speed of Vijay is $\frac{154}{9}$ m/s, which is equal to $\frac{154}{9} \times \frac{18}{5} = \frac{308}{5} = 61.6$ kmph.

The correct option is C

 VIDEO SOLUTION

 **100+ CAT DILR Questions**

